

Writing Growth and Decay Models from Data

Finding the multiplier from data, writing growth and decay models, and solving with them

Directions

- Every model on this sheet has the shape $y = a \cdot r^t$: a is the amount at $t = 0$ and r is the number you multiply by in each period.
 - From a table, r is the *ratio* of consecutive values. From a percent, $r = 1 + \frac{p}{100}$ for growth and $r = 1 - \frac{p}{100}$ for decay.
 - Round money, mass and population answers to two decimal places when a decimal appears; keep exact values otherwise.
 - State the units of t (hours, days, years) before you use a model, and keep them with your answer.
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1. A colony is counted every hour:

t (hours)	0	1	2	3
count	60	90	135	202.5

- (a) Find the hourly multiplier r . $r =$ _____

- (b) Use the model $60 \cdot r^t$ to predict the count at $t = 5$, to two decimal places. _____

- 2.** A quantity is measured once a year:

t (years)	0	1	2	3
amount	800	640	512	409.6

(a) Find the yearly multiplier r . $r =$ _____

(b) What percent of the amount is lost each year? _____

- 3.** A culture of 250 cells grows by 8% every hour.

(a) Write the hourly multiplier. $r =$ _____

(b) Using $250 \cdot r^t$, how many cells after 6 hours, to two decimal places? _____

- 4.** A car bought for \$18,000 loses 15% of its value every year.

(a) Write the yearly multiplier. $r =$ _____

(b) What is the car worth after 4 years, in dollars to two decimal places? _____

5. A sample weighs 800 mg at the start and 100 mg three hours later. Assume the mass follows $800 \cdot r^t$ with t in hours.

(a) Find the hourly multiplier r . $r =$ _____

(b) What is the mass after 5 hours? _____

6. A population of 300 doubles every 5 years, so after t years it is $300 \cdot 2^{t/5}$. After how many years does it reach 4800?

Answer: _____

7. A bacteria count is 40 on day 0 and 120 on day 1, and it keeps multiplying by the same factor each day.

(a) Find the daily multiplier. $r =$ _____

(b) What is the count on day 4? _____

8. A dose of 640 mg of a drug halves every hour, so the amount left after t hours is $640 \cdot \left(\frac{1}{2}\right)^t$ milligrams. After how many hours is 20 mg left?

Answer: _____ hr

9. A town of 12,000 people is losing 3% of its population each year.

(a) Write the yearly multiplier. $r =$ _____

(b) Using $12000 \cdot r^t$, what is the population after 10 years, to two decimal places? _____

10. A radioactive sample weighs 1200 g and its half-life is 8 days, so the mass after t days is $1200 \cdot \left(\frac{1}{2}\right)^{t/8}$ grams.

(a) What mass is left after 24 days?

Answer: _____ g

(b) After how many days is 37.5 g left? $t =$ _____

(c) Explain why halving the mass each half-life is not the same as removing a fixed number of grams each half-life.